

Lecture 11

- hypothalamus
- Limbic system
- Hippo Cam Pus
- Amygdala
- Cor Pus Callosum
- Neuro Transmitters & Behavior
- Effect of Hormones on CNS

A - Hypothalamus

- 1% of Brain mass
- Involve a relation between autonomic, endocrine & somatic functions
e.g. skeletal m. function

Important Parts:

^{lateral} Posterior Hypothalamus, Para Ventricular Nucleus, Medial preoptic area

Supra optic nucleus

Function

A - Cardio Vascular regulation

- Lateral posterior Hypothalamus → ↑ Heart R., ↑ B.P.
- Medial Pre optic area → ↓ Heart R., ↓ B.P.

B - Body Temperature Regulation

C - Body water Regulation (endocrine Function)

Supraoptic nucleus secrete ADH to regulate water level and electrolyte ~~Plan~~ Balance

D- uterine Contractility and milk ejection Regulation
 - Oxytocin is stored In Pituitary Gland and Released by Paraventricular Nucleus

- Oxytocin Regulate uterine Contraction and milk ejection occur as a neuro humoral reflex

E- Gastro Intestinal and Feeding Regulation

Low or High Glucose Levels Stimulate Hunger or Satiety Centers In Hypothalamus

F- Control anterior Pituitary Gland Function

- secrete ~~Releasing~~ ^{Releasing} Hormones that Release Pituitary Gland Hormones

- Releasing Hormones are

- Gonadotrophic RH (FSH, LH)

- GH-RH

- TSH-RH

- ACTH-RH

Hypothalamus Control 75% of Pituitary Gland Function

B- Limbic System

- border line between Hypothalamus, Related structure and Cerebral Cortex

- Controls animal emotional behavior

C- Hippo Campus

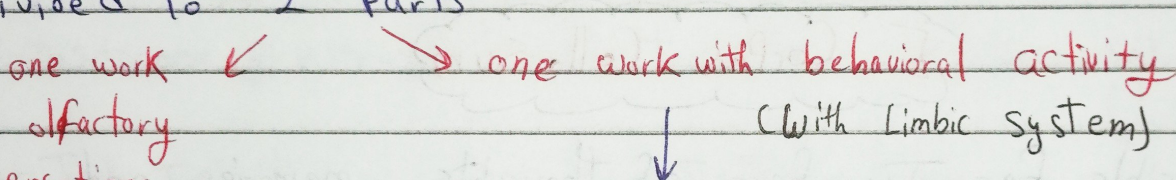
Limbic system Control behavior by signals Transmitted by Hippo Campus

- Stimulation ~~Damage or Hurting~~ of this Part Cause epileptic seizures and hallucination

D- Amygdala

- Amygdaloid nuclei found In Cerebral Cortex

- Divided To 2 Parts

one work with olfactory Sensations 

The Part receive signals from Limbic system, Hippo Campus & orbital Surface

- Amygdala is known as window through which Limbic system Realises The position of The creature In the world

- Amygdala Stimulation Leads To

- Inc. or Dec. In B.P / Heart R. / Gastro Intestinal secretion
- Pupil dilation / Pilo erection
- Defecation & micturation
- Pitutary Gland Hormones secretion

Other effects such as: Tonic movement, raising Head, body bending, Circular movements

Behavioral movement such as smell, eat, Lick, chew, swallow & sexual drive

- Destruction leads To ~~the~~ ~~the~~ Kluver Bucy Syndrome

Symptoms:

- No more fear
- forget too fast
- eat anything
- Inc. sex drive
- Copulate Immature animals or ~~the~~ wrong sex or different species

E. Corpus Callosum

- responsible for Transfere of thoughts, memories and Information between The Two hemispheres

Note: IF Left Hemisphere was damaged Right one Function will be affected

And The opposite occur IF Right one was The damaged one

F. Neuro Transmitters and behavior

Location of

- Nor-epinephrine Secreting Neurons:
 - reticular formation of locus Cereus of Limbic system
 - thalamus
 - cerebral Cortex
- Serotonin Secreting Neurons:
 - Pons & medulla (~~cerebrum~~) (Brain stem)
 - Limbic system

- Reduction in those neurotransmitters formation lead to Depression
signs: Grief, unhappiness, despair, misery, Insomnia,
loss of appetite & Increased Lust (sexual desire)

- Excess secretion especially of dopamine lead to Schizophrenia (~~plg~~) (plg)

G- Effect of Hormones on CNS

- Thyroid Hormone If Inc. leads to Increased ~~NS~~ NS activity and ~~met~~ metabolic Rate

- Testosterone & estrogen affect CNS and Inc. sexual desire (Libido)

- ACTH Injection leads to Intense fear

- Inc. activity of adrenal medulla leads to Inc. in brain activity and behavior changes